LECTURE 4

Biological File Formats

FASTA, FASTQ, GenBank, PDB, and GFF/GTF — structure, quality scores, and conversion

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Learning Objectives

1

Parse & Interpret

Read FASTA, FASTQ, GenBank, PDB, and GFF/GTF formats and identify what each field represents.

2

Explain

Describe how Phred quality scores are computed and what annotation fields in a record mean.

3

Convert

Move a record between common formats without losing essential information.

Topics Covered

1

Format Specifications

The exact structure and required fields of FASTA, FASTQ, GenBank, PDB, and GFF/GTF.

2

Phred Quality Scores

How a single ASCII character encodes a base-call error probability.

3

Annotation File Structure

How GFF/GTF describe genome features, and how GenBank embeds annotation with sequence.

Five Formats, Five Jobs



FASTA

Sequence only



FASTQ

Sequence + quality



GenBank

Sequence + rich annotation



PDB

3D atomic coordinates



GFF/GTF

Genome feature annotation

FASTA

FASTA

The simplest sequence format

```
>seq1 description here
ATGGCGTACCGTAGCTAGCTA
GCTAGCTAGCTAGCATCGATC
```

Header line

Starts with ">" — ID + optional description

Sequence line(s)

Wrapped at ~60-80 chars; no quality info

Multi-record

Multiple ">" blocks in one file (multi-FASTA)

FASTQ

FASTQ

Sequence + per-base quality — the sequencer's native output

```
@read1 machine:lane:tile
GGCTACGATCGTAGCTAGC
+
IIIIIHHGFEDCB@?>=<;
```

Line 1 (@)

Read identifier, mirrors FASTA header

Line 2

The nucleotide sequence

Line 3 (+)

Separator; may repeat the identifier

Line 4

One quality character per base (same length as line 2)

Phred Quality Scores

A single ASCII character encodes the probability that a base call is wrong.

$$Q = -10 \times \log_{10}(P)$$

Q = Phred score | P = probability the base call is incorrect

Phred Q	10	20	30	40
Error probability	1 in 10	1 in 100	1 in 1,000	1 in 10,000
ASCII char (Phred+33)	+	5	?	I

Phred+33 encoding: subtract 33 from the ASCII value of the quality character to get Q.

GB

GenBank Format

Sequence bundled with rich, structured annotation

```
LOCUS       NM_001301717   1200 bp
DEFINITION  Homo sapiens gene X mRNA
ACCESSION   NM_001301717
VERSION     NM_001301717.2
SOURCE      Homo sapiens (human)
FEATURES    Location/Qualifiers
  source     1..1200
  gene       1..1200
             /gene="GENEX"
  CDS        50..980
             /product="protein X"
ORIGIN
        1 atggcgtacc gtagctagct a...
//
```

LOCUS / DEFINITION

Name, length, and description

FEATURES

Annotated regions: gene, CDS, exon, etc.

ORIGIN

The raw sequence, numbered in blocks

//

Marks end of record

PDB

PDB Format

Fixed-width columns describing atomic coordinates

```
HEADER    OXYGEN TRANSPORT        22-JUL-84   4HHB
TITLE     THE CRYSTAL STRUCTURE OF HUMAN HEMOGLOBIN
ATOM      1  N   VAL A   1      6.204  16.869   4.854
ATOM      2  CA  VAL A   1      6.913  17.759   5.771
HETATM  145  FE  HEM A 142      8.128   9.596  -1.121
END
```

HEADER / TITLE

Metadata: classification, date, description

ATOM

One line per atom: name, residue, chain, x/y/z coords

HETATM

Non-standard groups (ligands, cofactors, waters)

END

Marks end of the coordinate entry

GFF

GFF / GTF Format

Tab-delimited genome feature annotation, 9 columns per line

seqid	source	type	start	end	score	strand	frame	attributes
chr1	ensembl	gene	1000	5000	.	+	.	ID=gene1

type — Feature kind: gene, mRNA, exon, CDS, etc.

start / end — 1-based inclusive genomic coordinates

strand — + (forward) or - (reverse)

attributes — Key=value pairs (GFF3) or key "value" (GTF) — IDs, parent links

GTF (Gene Transfer Format) is a stricter GFF2 variant, widely used by tools like Cufflinks and STAR.

Converting Between Formats

Conversion is easy when the target format holds a subset of the source's information — and lossy the other way.

FASTQ	→	**FASTA**	Straightforward — drop the quality line, keep header + sequence
GenBank	→	**FASTA**	Straightforward — keep ORIGIN sequence, discard FEATURES annotation
FASTA	→	**FASTQ**	Not directly possible — quality scores don't exist to invent
GFF	→	**GTF**	Mostly mechanical — reformat the attributes column syntax

Class Activities

1

Decode a Phred Quality String

In pairs, manually convert a short FASTQ quality string into Phred scores and error probabilities by hand, character by character.

2

Format-Conversion Relay

Each pair converts a record between two formats (e.g. FASTQ → FASTA) on paper, then hands it to the next pair to convert again — errors compound if a step is rushed.

Practical / Lab Work

Two tools, two languages — Biopython for sequence files, rtracklayer for annotation

PYTHON — Biopython SeqIO

```python
from Bio import SeqIO

for rec in SeqIO.parse("reads.fastq", "fastq"):
    q = rec.letter_annotations["phred_quality"]
    avg_q = sum(q) / len(q)
    print(rec.id, len(rec.seq), avg_q)
```

R — rtracklayer

```r
library(rtracklayer)

features <- import("annotation.gff3")
genes <- features[features$type == "gene"]
summary(width(genes))
```

Both parse the file into structured records so you never hand-split lines by delimiter — the library already knows each format's rules.

Assignment & Quick Check

A

Assignment

Write a parser that reports summary statistics — sequence length, GC%, and average quality score — for a given FASTA or FASTQ file.

Q

Quick Check for Understanding

Identify the file format from an unlabeled snippet:

`">seq1\nATGC..."`	`"@r1\nATGC\n+\nIIII"`	`"ATOM 1 N VAL..."`	`"chr1  ensembl gene..."`
FASTA	FASTQ	PDB	GFF/GTF